Microscopic–Macroscopic Independence and Poisson Peeling for Luce Permutations

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Abstract

We study the dependence between different cycle scales of Luce permutations. For rates converging uniformly to a positive continuous profile, every fixed vector of short-cycle counts is asymptotically independent of the entire ranked macroscopic PD(1) partition, with convergence of mixed factorial moments against bounded continuous partition observables. The proof verifies stability under finitely many prescribed cycles on good clock-slab data and then averages with factorial-moment weights. For arbitrary comparable-rate arrays, a marked switching argument gives a quantitative joint Poisson approximation for two separated cycle-length annuli with the remaining count as a passive mark. Its finite-depth error bounds can be iterated uniformly over logarithmic scales; fixed-moment transfer is justified separately from total variation. An appendix gives an analytic cutoff law, with an explicit error, for the number of parts of the limiting PD(1) partition. The main results concern joint dependence and complement, rather than reprove, the scalar total-cycle analytic mod-Poisson theorem.

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1 Model, scope, and main results

Chatterjee–Diaconis–Kim [4, Introduction] and Borga–Chatterjee–Diaconis [2, Section 5.2] ask for the cycle structure of Luce permutations, including joint cycle counts. Separate Poisson laws for short cycles and a Poisson–Dirichlet law for large cycles do not specify their dependence. The principal result here supplies this missing joint statement for regular positive profiles: every fixed short-cycle vector is asymptotically independent of the entire ranked macroscopic partition. This answers a part of the broader joint-cycle question; it does not claim universality for all positive-weight arrays.

Let $X_{n,i} \sim \mathrm{Exp}(\theta_{n,i})$, $1 \le i \le n$, be independent, and set

$$\sigma_n(i) = 1 + \#\{j : X_{n,j} < X_{n,i}\}.$$

This is the label-to-rank Luce permutation. Its inverse is the sequential weighted draw order and has the stationary Tsetlin-library distribution; inversion preserves all cycle counts. See Luce [11], Chatterjee–Diaconis–Kim [4], and Borga–Chatterjee–Diaconis [2]. Write $C_{n,r}$ for the number of r-cycles and $L_n(i)$ for the cycle length at label i. Put

$$K_n = \sum_{r=1}^{n} C_{n,r}, \qquad \mathbf{P}_n = (L_{n,1}/n, L_{n,2}/n, \ldots),$$

where $L_{n,1} \ge L_{n,2} \ge \cdots$ are the ranked cycle lengths. The partition space

$$\mathcal{S}^{\downarrow} = \{(x_j)_{j\ge1} : x_1 \ge x_2 \ge \cdots \ge 0,\ \sum_j x_j = 1\}$$

is equipped with the ℓ^1 metric.

We distinguish two hypotheses. The quantitative peeling results assume only a fixed bound on the largest-to-smallest rate ratio; after a common rescaling, write

$$1 \le \theta_{n,i} \le \varkappa < \infty. \tag{1.1}$$

For the short-cycle identification and joint microscopic–macroscopic limit, we assume

$$\theta_{n,i} = a_n\{f(i/n) + \varepsilon_{n,i}\}, \qquad \max_i |\varepsilon_{n,i}| \longrightarrow 0, \qquad f \in C[0,1], \quad 0 < \min f \le \max f < \infty. \tag{1.2}$$

The factor $a_n > 0$ rescales time only. In particular, (1.2) implies (1.1) with a fixed enlarged constant for all sufficiently large n.

Define

$$\mathcal{A}_f(t) = \int_0^1 (1 - e^{-f(x)t})\,dx, \qquad \mathcal{B}_f(t) = \mathcal{A}_f'(t), \qquad \tau_f = \mathcal{A}_f^{-1},$$

and the microscopic kernel and fixed-cycle means

$$\rho_f(x,y) = \frac{f(x)e^{-f(x)\tau_f(y)}}{\mathcal{B}_f(\tau_f(y))}, \qquad 0 < y < 1, \tag{1.3}$$

$$\lambda_r = \frac{1}{r}\int_{[0,1]^r} \rho_f(x_1,x_2)\cdots\rho_f(x_r,x_1)\,dx_1\cdots dx_r. \tag{1.4}$$

The cyclic integral is interpreted by increasing bulk truncations. Its finiteness and the fixed-cycle Poisson laws are companion-paper inputs, not new assertions of this paper. For falling factorials,

$(x)_j = x(x-1)\cdots(x-j+1)$ and $(x)_0 = 1$. We use d_{TV} for total variation (half the ℓ^1 distance on countable spaces). For a nonnegative integer variable with the required moments, κ_q^{F} denotes the factorial cumulant: the qth coefficient derivative of the formal series $\log \mathbb{E}(1+w)^Y$, or the corresponding analytic derivative when it exists. Only fixed orders are used in the peeling estimates.

Theorem 1.1 (Short cycles and the macroscopic partition)**.** *Under* (1.2)*, for every fixed* B*,*

$$(C_{n,1}, \dots, C_{n,B}, \mathbf{P}_n) \Longrightarrow (Z_1, \dots, Z_B, \mathbf{P})$$

in $\mathbb{N}_0^B \times \mathcal{S}^{\downarrow}$*, where* $Z_r \sim \mathrm{Pois}(\lambda_r)$ *are independent,* $\mathbf{P} \sim \mathrm{PD}(1)$*, and the Poisson vector and partition are independent. For every fixed nonnegative integer vector* $(k_1, \dots, k_B)$ *and bounded continuous* $F : \mathcal{S}^{\downarrow} \to \mathbb{R}$*,*

$$\mathbb{E}\left[F(\mathbf{P}_n) \prod_{r \le B} (C_{n,r})_{k_r}\right] \longrightarrow \mathbb{E}F(\mathbf{P}) \prod_{r \le B} \lambda_r^{k_r}. \tag{1.5}$$

The proof in Section 3 uses a uniform conditional macroscopic statement on good slab data. Merely combining the two marginal convergence theorems would not prove independence. A consequence is uniform mixed-moment convergence for macroscopic threshold counts on compact threshold intervals.

Our second result, Theorem 4.2, is a finite quantitative statement under (1.1). For nested cutoffs $b' < b < U < U'$, it compares the *joint* law of the lower-annulus count, the upper-annulus count, and the retained count with two independent Poisson variables independent of a copy of the retained count. The retained marginal is preserved exactly. Section 4 then gives stage-uniform iterated-logarithm estimates and explains the separate moment assumptions needed for unbounded tests.

Appendix A treats a separate property of the limiting PD(1) partition. It proves an analytic threshold-count limit with an explicit cutoff error by an elementary recurrence. This is neither a finite-n Luce convergence rate nor another proof of the scalar total-cycle theorem.

Relation to the scalar total-cycle paper. The profile-dependent mean shift, trace-class and Fredholm formula, universal factorial cumulants of K_n, and its local analytic mod-Poisson limit are proved in the companion manuscript [10]. We neither reprove these results nor treat analytic mod-Poisson convergence as an unresolved scalar problem. That manuscript is cited for context only: no result from it is a premise below. Conversely, its scalar transform alone does not imply the joint independence statement or the passive-mark estimates proved here. Quantitative total-cycle Poisson and lattice approximations are treated separately in [9].

2 Companion inputs and switching estimates

The named companion sources in the bibliography are pinned to a fixed repository revision. We state the interfaces used, including the conditional macroscopic information; unconditional PD(1) convergence alone would not be sufficient for Section 3.

2.1 Microscopic input

Input theorem 2.1 (Fixed short cycles)**.** *Under* (1.2)*, every fixed vector* $(C_{n,1}, \dots, C_{n,B})$ *converges to independent Poisson variables with means* (1.4)*, and every fixed mixed factorial moment converges to the corresponding Poisson moment.*

This is the regular-profile fixed-cycle theorem and mixed factorial-moment proposition of [6, Section 3]. That paper's local entry estimate is uniform in arbitrary distinct source labels; bulk and separation restrictions are imposed on the target ranks only. Its endpoint and collision envelopes justify the globally summed moments. We use the moment conclusion here, not a new marked local-entry argument.

2.2 Switching and a finite endpoint estimate

Assume (1.1). For $i \neq j$, put

$$R_{ij} = \exp\{-(\theta_{n,i} - \theta_{n,j})(X_{n,j} - X_{n,i})\}, \qquad R_{ii} = 1,$$

and define

$$S_{n,r} = \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}, \qquad e_{n,r} = \mathbb{E}|S_{n,r}/n - 1|, \qquad a_{n,r} = \frac{n}{r(n-r)}.$$

Here $L_n(i)$ is the length of the cycle containing i. Fix $p = 1 + 1/(4\varkappa)$ when $\varkappa > 1$, and take $p = 2$ when $\varkappa = 1$. Then $p < \varkappa/(\varkappa - 1)$ in the former case. Set $\beta = 1 - 1/p$; this choice makes all constants below depend only on the displayed parameters and on $\varkappa$.

Input theorem 2.2 (Marked switch and selected-factor control)**.** *For every bounded permutation-measurable F,*

$$r(n-r)\mathbb{E}[C_{n,r}F(\sigma_n)] = \mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}F\big(\sigma_n \circ (i\ \sigma_n^r(i))\big). \tag{2.1}$$

Moreover,

$$\sup_{n,r} \frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}^p < \infty. \tag{2.2}$$

The analogous p-moment bound holds for an independent uniform pair of labels. For such a pair (I, J), $\mathbb{E}R_{IJ} = 1$.

These are the marked master identity and selected-moment bounds of [5]. They immediately imply the following useful pointwise version of the short-root estimate.

Lemma 2.3 (Root point masses and exceptional switches)**.** *For $1 \le r < n$,*

$$\mathbb{E}C_{n,r} \le \frac{C_\varkappa n}{r(n-r)}. \tag{2.3}$$

Consequently, for an independent uniform label I_n and any set $E \subseteq \{1, \dots, \lfloor n/2 \rfloor\}$,

$$\mathbb{P}\{L_n(I_n) \in E\} \le C_\varkappa |E|/n. \tag{2.4}$$

For every set E of cycle lengths,

$$\frac{1}{n}\mathbb{E}\sum_{\substack{i:L_n(i)>r \\ L_n(i)\in E}} R_{i,\sigma_n^r(i)} \le C_\varkappa\big(\mathbb{P}\{L_n(I_n) \in E\}\big)^\beta. \tag{2.5}$$

Proof. Apply (2.1) with $F = 1$ and use (2.2) to bound $\mathbb{E}S_{n,r}$ by $C_\varkappa n$. Since

$$\mathbb{P}\{L_n(I_n) = r\} = r\mathbb{E}C_{n,r}/n,$$

(2.3) gives a bound $C_\varkappa/n$ for every $r \le n/2$. Summation proves (2.4). Hölder's inequality on the product of clock probability and the normalized counting measure on labels proves (2.5). □

We record the quantitative endpoint interface, including its truncation parameter. This avoids an appeal to an asymptotic admissibility condition at the terminal stages of the peeling argument. For a window $[r_-, r_+]$ with $r_+ \le \delta_0 n/2$, set

$$\delta = 2r_+/n, \qquad h = \delta^{1/4}, \qquad \omega = \delta^{1/(4\varkappa)}, \qquad M = n\sqrt{\delta}, \qquad L_\delta = 1 + \log(1/\delta).$$

The factor two accommodates the two-root construction. For an integer terminal depth d, define

$$\begin{aligned}\mathcal{B}_n(d; r_-, r_+) = (1-\alpha)^{d-1} + dh + d\omega + \frac{d^2 + d\log n}{M} + \omega + \delta + \delta^{1/2}L_\delta \\ + h^{-1}L_\delta e^{-cM} + n^2 h^{-1} e^{-chM} + n^{-2}.\end{aligned} \tag{2.6}$$

Here $c > 0$ and $\alpha \in (0,1)$ depend only on $\varkappa$.

Input theorem 2.4 (Finite truncation-uniform endpoint interface)**.** *There are positive constants c_0, δ_0 depending only on $\varkappa$ such that the following bounds hold for all sufficiently large n, provided*

$$1 \le d \le r_-/3, \qquad d \le c_0 hM, \qquad \delta \le \delta_0.$$

For every $T \ge 1$, put

$$Y^{(T)}_{n,r} = \frac{1}{n}\sum_{i: L_n(i) > r} (R_{i,\sigma^r_n(i)} \wedge T), \qquad D_{n,T} = \frac{1}{n^2}\sum_{i,j}(R_{ij} \wedge T).$$

Uniformly for $r \in [r_-, r_+]$,

$$|\mathbb{E}Y^{(T)}_{n,r} - \mathbb{E}D_{n,T}| \le C_\varkappa T \mathcal{B}_n, \tag{2.7}$$

$$|\mathbb{E}(Y^{(T)}_{n,r})^2 - \mathbb{E}D^2_{n,T}| \le C_\varkappa T^2 \mathcal{B}_n, \tag{2.8}$$

$$\operatorname{Var} D_{n,T} \le 4T^2/n. \tag{2.9}$$

The constants are independent of T, d, r_-, r_+.

This finite form is read from the estimates in the proof of the free-depth endpoint proposition (`prop:endpoint-mixing`, equation `eq:free-depth-global-bound`) and its observable corollary (`cor:quantitative-truncation-interface`) in [5]. These give (2.7)–(2.8). Their finite estimates use $d \le r_-/3$ to leave the frozen prefixes intact and $d \le c_0 hM$ to preserve residual row and cell margins. The conditions $d \to \infty$, $dh \to 0$, and $d\omega \to 0$ in the asymptotic formulation are used to make the displayed bound tend to zero, not to derive that bound. For large n, $M \ge \sqrt{2n}$ and $hM \ge 2^{3/4}n^{1/4}$, so the finite environment estimates apply uniformly. If the bracket is bounded below by a fixed positive constant, the first two estimates also follow from the trivial bounds $0 \le Y^{(T)}_{n,r}, D_{n,T} \le T$. The last inequality follows by replacing one clock: at most $2n$ summands change, each by at most T. No untruncated selected second moment is required.

Lemma 2.5 (Finite switching-intensity bound)**.** *Under the numerical conditions of Input theorem 2.4,*

$$\sup_{r_-\le r\le r_+} e_{n,r} \le C_\varkappa(\mathcal{B}_n(d;r_-,r_+)+n^{-1})^\eta, \qquad \eta=\frac{p-1}{2p}>0. \tag{2.10}$$

Proof. Write $\mu_{n,T}=\mathbb{E}D_{n,T}\le 1$. The selected and independent-pair p-moment bounds give

$$1-\mu_{n,T}\le C_\varkappa T^{1-p}, \qquad \mathbb{E}(S_{n,r}/n-Y_{n,r}^{(T)})\le C_\varkappa T^{1-p}.$$

Using (2.7)–(2.9),

$$\mathbb{E}|Y_{n,r}^{(T)}-\mu_{n,T}|^2\le C_\varkappa T^2(\mathcal{B}_n+n^{-1}).$$

Thus, with $u=\mathcal{B}_n+n^{-1}$,

$$e_{n,r}\le C_\varkappa\{T\sqrt{u}+T^{1-p}\}.$$

For $u\le 1$, choose $T=u^{-1/(2p)}$. For $u>1$, use the uniform first-moment bound on $S_{n,r}/n$. This proves the claim. □

2.3 Fixed-order cycle domination

Input theorem 2.6 (Fixed-order factorial domination)**.** *For every fixed $j\ge 1$, there is $C_{\varkappa,j}<\infty$ such that*

$$\mathbb{E}\mathcal{C}_n(r_1,\dots,r_j)\le\frac{C_{\varkappa,j}}{r_1\cdots r_j} \tag{2.11}$$

whenever $r_1+\cdots+r_j\le n/2$, where $\mathcal{C}_n(r_1,\dots,r_j)$ counts ordered tuples of distinct cycles of the specified lengths.

This is Corollary *Fixed-order factorial domination* (`cor:factorial-domination`) of [7], following its Exact fixed-order multi-clock identity (`thm:multiclock`), in the range where the total prescribed length is at most $n/2$. Only this finite-order probabilistic module is imported; no arithmetic order theorem or Edgeworth expansion is used.

2.4 Macroscopic input and its conditional scope

Input theorem 2.7 (Macroscopic partition and regular-history interfaces)**.** *Under* (1.1), $\mathbf{P}_n\Rightarrow$ PD(1) *in ℓ^1, and the size-biased residual deletion sequence converges to* GEM(1)*. Under* (1.2)*, we also use the following conditional interfaces of the same proof. Normalize clocks and rates by*

$$T_{n,i}=(\max_j\theta_{n,j})X_{n,i}, \qquad r_{n,i}=\theta_{n,i}/\max_j\theta_{n,j}\in[\varkappa^{-1},1].$$

For a fixed finite master partition of this normalized clock time, the complete slab data record the label sets H_a, the rank sets Q_a, and the ordered unlabeled *clock values. On good slab events of probability tending to one, each fixed row and cell population is bounded below by $c_{\mathcal{J}}n$. After conditioning only on compatible revealed edges and deleting complete cycles, the residual matchings are independent permanental matchings. The uniform rooted-closure and deletion estimates apply to regular residual histories with remaining mass at least a fixed positive fraction and relative row/cell error tending to zero. For a partition endpoint U, write $p_n(D(U))=n^{-1}\#\{i:T_{n,i}>U\}$ and*

$$\Delta_n(U,\delta,h)=\delta+2h+\chi_n+\sqrt{p_n(D(U))},$$

where h is the mesh of the finite clock cells. More precisely, if $S < U$ are fixed clock cutoffs, the interior root-CDF error is at most

$$C_{\varkappa,S,\xi,\zeta}\Delta_n(U,\delta,h) + C_\xi p_n(D(S)) + \pi_n + C_{\mathcal{J},\xi,\zeta}/n, \tag{2.12}$$

where ξ bounds the residual fraction below, ζ is the interior cutoff, and $\chi_n, \pi_n \to 0$ for the fixed partition. On the same good slab events, for a deterministic $\varepsilon_n \to 0$ and every partition endpoint U,

$$p_n(D(U)) \le e^{-U/\varkappa} + \varepsilon_n^2. \tag{2.13}$$

The deletion lemma is a joint-event *bound: the probability that an interior root cycle is deleted and the resulting history is not regular at inner cutoff S is at most π_n. The new regularity error is bounded by*

$$C_{\varkappa,S,\xi,\zeta}\Delta_n(U,\delta,h) + C_{\mathcal{J},\xi,\zeta}/n. \tag{2.14}$$

These bounds are uniform over regular starting histories. A uniform finite-dimensional deletion induction from a bounded prescribed deletion is derived below; it is not inferred from unconditional GEM(1) *convergence.*

The unconditional conclusions are Theorem 1.1 and Corollary 8.1 of [8]; the latter covers comparable arrays without an empirical rate-law limit. The conditional inputs are its residual-factorization lemma, uniform residual-root law, and deletion-stability lemma. The explicit bounds above are its equations `eq:root-cdf` and `eq:deletion-error`, together with the joint-event assertion `eq:deletion-joint`. Under (1.2), the normalized empirical rate law converges to the law ν of $f(U)/\max f$, with U uniform on $[0,1]$. This law is supported on $[\varkappa^{-1}, 1]$. The common-component argument of Corollary 8.1 gives $q(\nu) \le \sqrt{1-\varkappa^{-1}} < 1$, and the channel density satisfies $B_\nu(S) = \int re^{-rS}\nu(\mathrm{d}r) \ge \varkappa^{-1}e^{-S}$. Thus the conditional estimates apply with fixed finite constants. The tail bound (2.13) follows from $r_{n,i} \ge \varkappa^{-1}$ and concentration of the empirical clock tails at the finitely many endpoints; the good slab event is intersected with these events. No uniformity over varying profile functions is needed. The regularity tested there compares, on the rows below the chosen outer cutoff,

$$\frac{|H_a \cap R|}{v_R|H_a|}, \qquad \frac{|Q_a \cap H_b \cap R|}{v_R|Q_a \cap H_b|}, \qquad v_R = |R|/n,$$

with one, uniformly in the target category b. The histories contain no additional information about unexposed edges. The source labels `lem:residual-factorization`, `prop:uniform-root-fraction`, `lem:deletion-stability` identify the conditional modules. The cutoff ordering follows the proof of `prop:residual-uniforms`, but its unconditional statement is not used as a substitute for the displayed conditional estimates. Section 3 verifies these conditions for every compatible collection of finitely many prescribed complete cycles before invoking their uniformity. No bound on the probability of a bad slab event *conditional on a rare prescribed cycle* is assumed.

3 Microscopic–macroscopic independence

Throughout this section assume (1.2). We first establish the finite-deletion consequence of Input 2.7 that is needed for weighted averaging. All prescribed cycles in this section have bounded total size, fixed before $n \to \infty$.

3.1 Uniformity after finitely many prescribed cycles

Lemma 3.1 (Finite-deletion law on good slab data)**.** *Fix $v < \infty$ and a bounded Lipschitz function F of the ranked partition in the ℓ^1 metric. Let $c_F = \mathbb{E}F(\mathbf{P})$ for $\mathbf{P} \sim \mathrm{PD}(1)$. For every $\eta > 0$ there are a fixed finite clock partition, its complete slab-data sigma-field $\mathscr{S}_n$, and $\mathscr{S}_n$-measurable events $\mathcal{G}_n$ with $\mathbb{P}(\mathcal{G}_n^c) \to 0$ such that*

$$\limsup_{n\to\infty} \operatorname*{ess\,sup}_{\mathscr{S}_n \in \mathcal{G}_n} \sup_{\substack{\Gamma \text{ compatible} \\ |V(\Gamma)|\le v}} |\mathbb{E}[F(\mathbf{P}_n) \mid \mathscr{S}_n, \Gamma] - c_F| \le \eta. \tag{3.1}$$

Here Γ is a prescribed collection of disjoint complete cycles and conditioning on Γ means conditioning on their specified edges. Only compatible collections of positive conditional probability are used.

Proof. Use the complete slab data and regular-history interfaces of Input 2.7: the label sets H_a, the rank intervals Q_a, and the ordered *unlabeled* clock values within each slab. Given these data, the within-slab matchings are independent permanental matchings. Conditioning on compatible specified edges removes their sources and targets and leaves the same permanental laws on the residual minors. This is the residual-factorization lemma of that paper, and does not condition on the fully labeled clocks.

Write $m_a = |H_a|$, $N_{ab} = |Q_a \cap H_b|$, and let R be the vertex set remaining after deletion of Γ. Put $r = |R|/n$, $m_a^R = |H_a \cap R|$, and $N_{ab}^R = |Q_a \cap H_b \cap R|$. For each deleted complete cycle C,

$$|C \cap H_a| = |C \cap Q_a|,$$

because σ_n maps the first set bijectively onto the second. Thus the residual rows remain square. For the fixed partition, the good slab event in [8] has every row and cell population at least $c_{\mathcal{J}} n$. Since deletion changes each population by at most v,

$$\max_a \left| \frac{m_a^R}{r m_a} - 1 \right| + \max_{a,b} \left| \frac{N_{ab}^R}{r N_{ab}} - 1 \right| \le C_{\mathcal{J}} v/n. \tag{3.2}$$

This verifies the companion's regular-history hypothesis uniformly over Γ.

Fix the number k of further uniform residual-root deletions, an interior cutoff $\zeta \in (0, 1/4)$, and a desired finite-dimensional error. For all large n, the initial residual size is at least $n/2$; on the first k interior deletions it stays at least $\xi_* n$, where $\xi_* := \zeta^k/2$. Choose clock cutoffs $T_0 < \cdots < T_k$ from inner to outer, and then a single master partition with mesh h containing those cutoffs. At stage $j+1$, use outer cutoff T_{k-j} and inner cutoff T_{k-j-1}. At each inner cutoff T_i, let R_i and B_i bound respectively the principal root-law and deletion coefficients, with residual fraction at least ξ_*, and set

$$A_i = \prod_{\ell=0}^{i} (1 + R_\ell + B_\ell).$$

All these constants are fixed before T_{i+1} is chosen. A tail error introduced at level j enters a later regularity bound with coefficient $B_i \cdots B_j$, or a later root-law bound with coefficient $R_i B_{i+1} \cdots B_j$, for some $i \le j$; both are at most A_j. Choose T_0 to control the direct root tail, and each T_{i+1} so that $A_i e^{-T_{i+1}/(2\varkappa)}$ is smaller than its allocated error budget. Only after this finite hierarchy is fixed is h chosen so that $2h \sum_{i<k} A_i$ is small. Finally take n large enough to control the fixed-partition errors and their finitely many amplification factors.

Here is the explicit perturbation from a prescribed deletion. Put $d_0 = C_{\mathcal{J}} v/n$ and, for $0 \le j < k$, define

$$d_{j+1} = B_{k-j-1}(d_j + 2h + \chi_n + \varepsilon_n + e^{-T_{k-j}/(2\varkappa)}) + C_{\mathcal{J},k,\xi_*,\zeta}/n, \tag{3.3}$$

where B_i bounds the deletion coefficient with inner cutoff T_i. All sequences are replaced by the maxima over the finitely many cutoff levels. Relative to the same recursion starting from zero, the only additional error is

$$d_j - d_j^{(0)} = \frac{C_{\mathcal{J}} v}{n} \prod_{i=0}^{j-1} B_{k-i-1} = o_n(1),$$

uniformly in the compatible prescribed cycles, since all constants are fixed before $n \to \infty$. The hierarchy makes $d_j < 1/4$ throughout. Equations (2.12)–(2.14) then apply at every stage with these deterministic regularity bounds.

To pass from these conditional bounds to a joint law, first use bounded Lipschitz one-variable tests supported in $[\zeta, 1-\zeta]$ and vanishing at the endpoints. Conditional Stieltjes integration bounds their expectation errors by the interior CDF error times the integral of the absolute derivative. Multiply by the previously exposed tests and iterate. The joint-event deletion bound charges failures of the next regular history directly; it never conditions on the exact new cycle length. The total failure probability is at most the sum of the k uniform failure bounds. Evaluating the CDF at 2ζ and $1 - 2\zeta$ bounds the omitted endpoint probabilities by $4k\zeta$ plus these errors. Letting the chosen finite-dimensional error and then ζ tend to zero proves the product-uniform comparison, uniformly over the good initial slab data and the prescribed cycles. Products of Lipschitz functions and a finite uniform approximation on $[0,1]^k$ give the comparison for any fixed continuous test of the first k residual fractions.

In particular, for any prescribed $\eta_1 > 0$ the hierarchy can be chosen so that the residual mass R_k^{res}, normalized by the initial residual size $|R|$, satisfies

$$\mathbb{E}[R_k^{\mathrm{res}} \mid \mathscr{S}_n, \Gamma] \le 2^{-k} + \eta_1 + o_n(1)$$

uniformly: apply the finite-dimensional comparison to $\prod_{j=1}^{k}(1 - u_j)$. A bounded Lipschitz test on the ranked partition can therefore be reduced to its first k size-biased pieces at cost $C_F(2^{-k} + \eta_1 + o_n(1))$; decreasing rearrangement is a contraction in ℓ^1. Formally, extend F to the ambient ℓ^1 space by a bounded Lipschitz extension before truncating the sequence.

Finally, restoring the deleted pieces and changing normalization from $|R|$ to n changes the ranked sequence by at most $2v/n$ in ℓ^1. Choose k large and then the finite-dimensional errors small enough for the prescribed η. This proves (3.1). The regular-profile hypothesis ensures convergence of the marked empirical rate measures, as required for this fixed-partition construction. □

Proof of Theorem 1.1. Set

$$W_n = \prod_{r \le B} (C_{n,r})_{k_r}, \qquad v = \sum_{r \le B} r k_r.$$

Expand W_n as a sum of indicators over ordered collections Γ of the prescribed disjoint cycles. The total number of exposed vertices is v, independent of n. By Input theorem 2.1, $\mathbb{E}W_n$ converges to $\prod_r \lambda_r^{k_r}$. Also $\sup_n \mathbb{E}W_n^2 < \infty$: the square is a fixed polynomial in falling factorials of the finitely many short-cycle counts, whose mixed factorial moments converge by Input 2.1. This part of the proof does not require fixed-order domination for growing cycle lengths.

First let F be bounded Lipschitz. Apply Lemma 3.1 with error η. On $\mathcal{G}_n$, condition first on the complete slab data and then on one prescribed collection. Sum the resulting inequalities with their conditional occurrence probabilities. On the bad slab event use Cauchy–Schwarz. This gives

$$\begin{aligned}|\mathbb{E}[F(\mathbf{P}_n)W_n] - c_F\mathbb{E}W_n| &\le (\eta + o_n(1))\mathbb{E}W_n \\ &\quad + 2\|F\|_\infty(\mathbb{E}W_n^2)^{1/2}\mathbb{P}(\mathcal{G}_n^c)^{1/2}.\end{aligned} \tag{3.4}$$

Let $n \to \infty$ and then $\eta \downarrow 0$. This proves the mixed factorial-moment factorization for bounded Lipschitz tests. It never requires a uniform bound on $\mathbb{P}(\mathcal{G}_n^c \mid \Gamma)$.

For completeness, this factorization determines joint weak convergence. Both marginals are tight. Along any joint subsequential limit, uniform integrability transfers the mixed factorial moments; conditioning by a nonnegative bounded Lipschitz test of the partition leaves the same factorial moments as the independent Poisson vector. The Poisson vector is moment-determinate. For example, its factorial moments give a finite exponential moment by the monotone expansion of $\prod_r (1+t_r)^{Z_r}$ for $t_r \ge 0$. Thus every subsequential limit is the stated product law. Finally, joint weak convergence and the uniform integrability of W_n extend (1.5) to bounded continuous F. □

Corollary 3.2 (Uniform threshold moments)**.** *Fix B, integers $k_1, \dots, k_B, k \ge 0$, and $[a,b] \subset (0,1)$. Define*

$$N_{n,t} = \sum_{r > tn} C_{n,r}, \qquad N_t = \#\{j : P_j > t\}, \qquad \mathbf{P} \sim \mathrm{PD}(1).$$

Then, uniformly for $t \in [a,b]$,

$$\mathbb{E}\left[\prod_{r \le B} (C_{n,r})_{k_r} (N_{n,t})_k\right] \longrightarrow \left(\prod_{r \le B} \lambda_r^{k_r}\right) \mathbb{E}(N_t)_k. \tag{3.5}$$

Consequently the sum $\sum_{r \le B} C_{n,r} + N_{n,t}$ has the same limiting fixed factorial moments, uniformly on $[a,b]$, as $P_B + N_t$, where $P_B \sim \mathrm{Pois}(\sum_{r \le B} \lambda_r)$ is independent of N_t.

Proof. For a fixed $t > 0$, a PD(1) partition has no part exactly equal to t, and the threshold-count map is continuous at every such partition in ℓ^1. Moreover, $N_{n,t} \le 1/a$ for $t \in [a,b]$. Use Theorem 1.1 and uniform integrability of the short-cycle factor to obtain pointwise convergence in (3.5).

For fixed short-cycle factorial factor $W_n \ge 0$, the functions $g_n(t) = \mathbb{E}[W_n(N_{n,t})_k]$ are non-increasing. Their limit is continuous on $[a,b]$: the PD factorial moment measure, recalled in Lemma A.2, gives continuity by dominated convergence on $[a,1]^k$. A finite-grid monotonicity sandwich therefore upgrades pointwise to uniform convergence. Expand the factorial moments of the sum by the binomial identity for falling factorials. The same conclusion holds simultaneously for any fixed finite set of B's. □

4 Annular passive-Poisson peeling

For integers $b < U$, define

$$D_n(b,U) = \sum_{r \le b} C_{n,r} + \sum_{r > U} C_{n,r}, \qquad H(a,b) = \sum_{a < r \le b} \frac{1}{r}.$$

All moment orders in this section are fixed before the cutoffs and error exponents are chosen. The quantity $e_{n,r} = \mathbb{E}|S_{n,r}/n - 1|$ is an absolute switching-intensity error. A bound only on $|\mathbb{E}(S_{n,r}/n) - 1|$, sufficient for scalar real-fugacity growth, would not control the passive-mark test functions used here.

4.1 Marked Stein comparison and annular errors

Lemma 4.1 (Passive-mark Stein transfer)**.** *Let X be a nonnegative integer-valued variable with finite mean, let D be a countable-valued mark, and let $\lambda \geq 0$. Suppose that for every bounded g with $\|g\|_\infty \leq 1$,*

$$|\mathbb{E}[Xg(X-1,D)] - \lambda \mathbb{E}g(X,D)| \leq \eta,$$

where the term with $X = 0$ is defined as zero. Then

$$d_{\mathrm{TV}}(\mathcal{L}(X,D), \mathrm{Pois}(\lambda) \otimes \mathcal{L}(D)) \leq \eta.$$

Proof. Use the immigration–death generator on the active count while keeping the mark fixed. For an event B, center its indicator by its Poisson average at the same mark. The semigroup solution h_B has $|h_B(x+1,d) - h_B(x,d)| \leq 1$: couple chains differing by one initial particle, which survives until time t with probability e^{-t}, and integrate over $t \geq 0$. Testing the assumed Campbell discrepancy with $g(x,d) = h_B(x+1,d) - h_B(x,d)$ bounds the difference of event probabilities by η. The centering integrates to $(\mathrm{Pois}(\lambda) \otimes \mathcal{L}(D))(B)$, so the mark retains its original marginal. This is the marked Poisson Stein argument; see [1, 9]. □

Theorem 4.2 (Joint annular peeling)**.** *Assume* (1.1)*. Let positive integer cutoffs satisfy*

$$b' < b < U < U' < n/4, \qquad U + b < n/2.$$

Put

$$J^- = \sum_{b' < r \leq b} C_{n,r}, \qquad J^+ = \sum_{U < r \leq U'} C_{n,r}, \qquad \Delta(I) = \sum_{r \in I} a_{n,r} e_{n,r}.$$

Let

$$\lambda^- = \sum_{b' < r \leq b} a_{n,r}, \qquad \lambda^+ = \sum_{U < r \leq U'} a_{n,r}.$$

There are independent $Z^- \sim \mathrm{Pois}(\lambda^-)$, $Z^+ \sim \mathrm{Pois}(\lambda^+)$ and $\widetilde{D}_n(b',U') \stackrel{d}{=} D_n(b',U')$ such that

$$\begin{aligned} &d_{\mathrm{TV}}(\mathcal{L}(J^-, J^+, D_n(b',U')), \mathcal{L}(Z^-, Z^+, \widetilde{D}_n(b',U'))) \\ &\quad \leq C_\varkappa \left\{ \Delta((b',b]) + \Delta((U,U']) + H(b',b)(b/n)^\beta + H(U,U')(U'/n)^\beta \right\}. \end{aligned} \tag{4.1}$$

Proof. For the lower annulus use the countable-valued vector $M^- = (J^+, D_n(b',U'))$ as passive mark. A split at distance $r \in (b',b]$ increases J^- by one and leaves M^- unchanged whenever its source-cycle length is outside

$$E_r^- = (r, r+b] \cup (U, U+r] \cup (U', U'+r].$$

Indeed, outside the first interval both the source and residual cycles are longer than b; the other two exclusions keep them on the same sides of both U and U'. The U' exclusion is needed because we retain the upper-annulus count and the final remainder separately. The set E_r^- has at most $b + 2r \leq 3b$ integer points and is contained in $\{1, \ldots, \lfloor n/2 \rfloor\}$: its largest possible endpoint is $U' + r < U' + b < 2U' < n/2$. For instance, $b' = 2, b = 5, U = 12, U' = 20, r = 3$ and source length 22 show why the last interval cannot be omitted: the split creates the upper-annulus cycle of length 19 and removes a long retained cycle. This witness is admissible for any $n > 80$.

Apply (2.1) with the test $g(J^-(\pi)-1, M^-(\pi))$ on permutations with $J^-(\pi) > 0$, and zero otherwise. The discrepancy from the unchanged test is at most two on exceptional splits. Thus, for $\|g\|_\infty \le 1$,

$$\begin{aligned}
&\bigl|\mathbb{E}[C_{n,r}\, g(J^- - 1, M^-)] - a_{n,r}\mathbb{E}\, g(J^-, M^-)\bigr| \\
&\quad \le a_{n,r} e_{n,r} + \frac{2a_{n,r}}{n}\mathbb{E} \sum_{\substack{i: L_n(i) > r \\ L_n(i) \in E_r^-}} R_{i,\sigma_n^r(i)} \\
&\quad \le a_{n,r} e_{n,r} + C_\varkappa a_{n,r} (b/n)^\beta,
\end{aligned}$$

where the last step is Lemma 2.3. Summing over r and applying Lemma 4.1 gives a joint approximation of (J^-, M^-) by $\mathrm{Pois}(\lambda^-) \otimes \mathcal{L}(M^-)$ with the first and third errors in (4.1); here $a_{n,r} \le 2/r$.

Next apply the same argument to J^+ with passive mark $D_n(b', U')$. For $r \in (U, U']$, the split preserves this mark and adds one to J^+ whenever its source length exceeds $r + U'$. The sufficient exceptional set $(r, r+U']$ has at most U' points and lies below $n/2$, since $r \le U'$ and $2U' < n/2$. It is deliberately conservative: no assertion is needed about splits inside that set. The same argument gives the second and fourth errors, with parameter λ^+. Tensoring this second approximation with the independent Z^- and applying the triangle inequality proves (4.1). Applying the summation map also gives the corresponding comparison of $D_n(b, U)$ with $Z^- + Z^+ + \widetilde{D}_n(b', U')$. □

4.2 Moment bounds and factorial-cumulant transfer

Lemma 4.3 (Harmonic moment bound)**.** *Fix $k \ge 1$ and set $W = \lfloor n/(2k) \rfloor$. For any set I of cycle lengths, let $V_I = \sum_{r \in I} C_{n,r}$ and $h_I = \sum_{r \in I,\ r \le W} 1/r$. Then, for all sufficiently large n,*

$$\mathbb{E} V_I^k \le C_{\varkappa,k}(1 + h_I)^k. \tag{4.2}$$

In particular, $\mathbb{E} K_n^k \le C_{\varkappa,k}(1 + \log n)^k$.

Proof. Let $V = \sum_{r \in I,\ r \le W} C_{n,r}$. For $1 \le j \le k$, expansion into ordered distinct-cycle tuples and Input theorem 2.6 give

$$\mathbb{E}(V)_j \le C_{\varkappa,j} h_I^j,$$

because each tuple has total length at most $jW \le n/2$. The expansion of V^k in falling factorials proves the corresponding ordinary-moment bound. There are fewer than $2k$ cycles longer than W, since each such cycle has length greater than $n/(2k)$. Adding this bounded count proves (4.2). □

Lemma 4.4 (Factorial-cumulant transfer)**.** *Fix $q \ge 2$. If X, Y are nonnegative integer-valued variables with*

$$d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(Y)) \le \eta, \qquad \mathbb{E}X^{2q} + \mathbb{E}Y^{2q} \le C_q H^{2q}, \qquad H \ge 1,$$

then

$$\bigl|\kappa_q^{\mathrm{F}}(X) - \kappa_q^{\mathrm{F}}(Y)\bigr| \le C_q' H^q \eta^{1/2}. \tag{4.3}$$

Proof. Under a maximal coupling, $\mathbb{P}(X \ne Y) \le \eta$. For $1 \le j \le q$, Cauchy–Schwarz and the $2q$-th-moment bound imply

$$|\mathbb{E}(X)_j - \mathbb{E}(Y)_j| \le \mathbb{E}[(X^j + Y^j)\mathbf{1}_{\{X \ne Y\}}] \le C_{q,j} H^j \eta^{1/2}.$$

Writing $\mathcal{P}_q$ for the set of partitions of $\{1,\dots,q\}$, the factorial cumulant is the fixed polynomial

$$\kappa_q^{\mathrm{F}}(X) = \sum_{\pi\in\mathcal{P}_q} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} \mathbb{E}(X)_{|B|}.$$

Telescoping each product and using $\sum_{B\in\pi}|B| = q$ proves the claim. The constants depend on the fixed order; no uniformity in q is asserted. □

4.3 Uniform iterated-logarithm estimates

Define successive logarithms while they are positive by

$$\ell_1 = \log n, \qquad \ell_{j+1} = \log \ell_j,$$

and, for a fixed exponent A,

$$b_j = \lceil \ell_j^3 \rceil, \qquad U_j = \lfloor n/\ell_j^A \rfloor, \qquad D_j = D_n(b_j, U_j).$$

Lemma 4.5 (Iterated-log peeling estimate)**.** *Fix $q \ge 2$ and $Q > 2q+4$. One can choose A sufficiently large so that, for some H_0 and all sufficiently large n, the following assertions hold simultaneously at every stage with $x := \ell_{j+1} \ge H_0$:*

$$d_{\mathrm{TV}}(\mathcal{L}(D_j), \mathcal{L}(Z_j^- + Z_j^+ + \widetilde{D}_{j+1})) \le C_{\varkappa,Q,A} x^{-Q}, \tag{4.4}$$

$$\mathbb{E}D_j^{2q} + \mathbb{E}(Z_j^- + Z_j^+ + \widetilde{D}_{j+1})^{2q} \le C_{\varkappa,q,A}(1+x)^{2q}, \tag{4.5}$$

where $Z_j^-, Z_j^+, \widetilde{D}_{j+1}$ are independent, $\widetilde{D}_{j+1} \overset{d}{=} D_{j+1}$, and the Poisson parameters are

$$\lambda_j^- = \sum_{b_{j+1}<r\le b_j} a_{n,r}, \qquad \lambda_j^+ = \sum_{U_j<r\le U_{j+1}} a_{n,r}.$$

Consequently,

$$\left|\kappa_q^{\mathrm{F}}(D_j) - \kappa_q^{\mathrm{F}}(D_{j+1})\right| \le C_{\varkappa,q,Q,A} x^{q-Q/2}. \tag{4.6}$$

Proof. Since $\ell_j = e^x$, harmonic summation gives

$$\begin{aligned} H(b_{j+1}, b_j) &= 3(x - \log x) + O(1), \\ H(U_j, U_{j+1}) &= A(x - \log x) + O(1). \end{aligned} \tag{4.7}$$

In particular, the harmonic mass of either annulus is $O_A(x)$. For large H_0 and then large n, all cutoff inequalities in Theorem 4.2 hold uniformly. Indeed, $b_j \le 2(\log n)^3$, $U_j \ge n/(2(\log n)^A)$, and $U_{j+1}/n \le x^{-A}$.

Choose a fixed $D > 0$ and terminal depth $d = \lceil D \log x \rceil$. On the lower annulus, $r_- \ge b_{j+1} \ge x^3$; on the upper annulus, $r_- \ge U_j$. Thus $d \le r_-/3$ once H_0 is large. Also $d = O_D(\log n)$ and $hM \ge cn^{1/4}$, so the numerical residual margin condition in Input theorem 2.4 holds uniformly.

For clarity, retain the finite-n error in the calculation. The reset term is at most Cx^{-cD}. For the lower annulus, its largest relative length is $O((\log n)^3/n)$; hence all nonreset terms in (2.6), after applying Lemma 2.5, are bounded by $n^{-c_1}(\log n)^{C_1}$ for some $c_1 > 0$. For the upper annulus, its largest relative length is $O(x^{-A})$. The terms $dh, d\rho, \rho, \delta, \delta^{1/2}L_\delta$ are bounded by

$$C_{\varkappa,A,D}(1+\log x)^C x^{-a_0 A}$$

for some $a_0 > 0$ after the same fixed positive power is applied. The terms $(d^2 + d\log n)/M$ and the environment failures again contribute at most $n^{-c_1}(\log n)^{C_1}$, after changing constants. The selected-factor exceptional terms are bounded using $(b_j/n)^\beta$ and $x^{-A\beta}$.

Thus Theorem 4.2 and (4.7) give

$$\begin{aligned}\text{error in (4.4)} \le C_{\varkappa,A,D}x\big[x^{-c_2D} + (1+\log x)^{C_2}x^{-a_0A} + x^{-A\beta}\big] \\ + C_{\varkappa,A,D}x\, n^{-c_1}(\log n)^{C_1}.\end{aligned} \tag{4.8}$$

Choose D with $c_2D > Q+3$, and then A with $\min(a_0,\beta)A > Q+3$. Since every stage has $x \le \log\log n$, the last term is $o(x^{-Q})$ uniformly. The spare powers absorb the logarithmic factors. This proves (4.4); constants are independent of the number of stages.

For the moments use Lemma 4.3 with $k = 2q$ and $W = \lfloor n/(4q)\rfloor$. For the lengths contributing to D_j,

$$\sum_{r\le b_j}\frac{1}{r} + \sum_{U_j<r\le W}\frac{1}{r} \le (A+3)x + O_q(1).$$

The same bound holds for D_{j+1}, since its corresponding scale is $\log x \le x$. Finally, by $a_{n,r} \le 2/r$ and (4.7), $\lambda_j^- + \lambda_j^+ = O_A(x)$. Fixed Poisson moments are polynomials in the mean, which proves (4.5). Apply Lemma 4.4 and additivity under independence. Both Poisson factorial cumulants vanish for $q \ge 2$, giving (4.6). □

Corollary 4.6 (Accumulated peeling error)**.** *Fix the parameters of Lemma 4.5 and $H \ge H_0$. For all sufficiently large n, let $J = J(n,H)$ be the last index with $\ell_J \ge H$. There are independent Poisson variables with means λ_j^-, λ_j^+, $1 \le j < J$, independent of a variable $\widetilde{D}_J \overset{d}{=} D_J$, such that*

$$d_{\mathrm{TV}}\left(\mathcal{L}(D_1), \mathcal{L}\left(\sum_{j<J}(Z_j^- + Z_j^+) + \widetilde{D}_J\right)\right) \le C_{\varkappa,Q,A}H^{-Q}. \tag{4.9}$$

For the fixed factorial-cumulant order q of that lemma,

$$|\kappa_q^{\mathrm{F}}(D_1) - \kappa_q^{\mathrm{F}}(D_J)| \le C_{\varkappa,q,Q,A}H^{q-Q/2}. \tag{4.10}$$

Proof. Iterate the scalar consequence of Theorem 4.2; convolution with a fixed probability law contracts total variation. At every step use independent copies in the comparison law. The total error is bounded by $C\sum_{j<J}\ell_{j+1}^{-Q}$. Read this finite sequence backward. Successive values are related by $x \mapsto e^x$ and, for H_0 large, are at least twice their predecessors. Thus the sum is at most CH^{-Q}, with a constant independent of J. The same argument applied to (4.6), using $q - Q/2 < 0$, proves the second bound. □

Remark 4.7 (Meaning of the iterated comparison). Equation (4.9) is a comparison of the scalar retained count with a convolution law. It does not assert simultaneous independence of all actual annular counts at all stages. The one-stage vector estimate in Theorem 4.2 is stronger than its scalar consequence, but iteration with additional retained coordinates would require preserving those coordinates in every switching test. Neither the number of factorial-cumulant orders nor the number of arbitrary passive coordinates is allowed to grow under the fixed-order statements above.

5 Scope and further questions

The independence theorem concerns every fixed short-cycle vector and the whole ranked partition; the quantitative peeling estimates concern unweighted comparable-rate arrays. The explicit cutoff error in Theorem A.1 belongs to the limiting PD(1) model. It is not a convergence rate for $\mathbf{P}_n$, nor does it justify interchanging a moving microscopic cutoff with $n \to \infty$.

A natural extension is to allow the number of observed short-cycle lengths or the size of the prescribed deletion to grow. This would require quantitative microscopic moment bounds and a residual-history estimate uniform in that growing size. Another question is a joint total-cycle and marked short-cycle generating-function limit under a fixed real or complex cycle tilt. The scalar result of [10] and the untilted joint limit here do not, by themselves, supply such a theorem: one must also control the tilted dependence and unbounded generating tests.

The positive-profile assumption in the independence theorem excludes endpoint-singular families. Extending these joint statements to such families requires a separate boundary-process analysis; no transfer is asserted from comparable rates alone when the rate ratio is unbounded.

Dependency boundary. The independence theorem uses the microscopic mixed factorial moments and the conditional regular-history interfaces of the macroscopic paper. The annular and iterated comparisons additionally use the exact marked switch, finite two-root endpoint interface, and the fixed-order factorial domination stated in Section 2. Their quantitative scope is recorded there. The PD(1) cutoff argument uses the standard size-biased representation and the cited Dickman density. The scalar mean-shift, Fredholm, and total-cycle cumulant arguments from the earlier manuscript are not dependencies and are not reproduced. The present draft has been internally checked against these interfaces; no claim of independent external refereeing or formal verification is made.

A An analytic cutoff law for the limiting PD partition

Let $\mathbf{P} = (P_1, P_2, \ldots) \sim \mathrm{PD}(1)$ and $N_\varepsilon = \#\{j : P_j > \varepsilon\}$.

This appendix is independent of the Luce conditioning and switching arguments. Its error is a function of the cutoff in the limiting partition, not of the finite permutation size.

Theorem A.1 (Analytic cutoff law for PD(1))**.** *Let* $\mathbf{P} \sim \mathrm{PD}(1)$ *and* $N_\varepsilon = \#\{j : P_j > \varepsilon\}$*. For every* $0 < \varrho < 1/2$*, uniformly for complex* $|w| \le \varrho$*,*

$$\varepsilon^w \mathbb{E}(1+w)^{N_\varepsilon} = \frac{e^{-\gamma w}}{\Gamma(1+w)} + O_\varrho(\varepsilon^{1-2\varrho}) \qquad (\varepsilon \downarrow 0). \tag{A.1}$$

Here $\varepsilon^w = \exp(w \log \varepsilon)$ *uses the real logarithm, and* γ *is Euler's constant.*

Lemma A.2 (Factorial moment measures of PD(1))**.** *For every* $k \ge 1$*, the ordered* k*-point factorial moment measure of the parts is*

$$\mathbf{1}_{\{x_1+\cdots+x_k<1\}} \frac{\mathrm{d}x_1 \cdots \mathrm{d}x_k}{x_1 \cdots x_k}. \tag{A.2}$$

Proof. Successive size-biased sampling of the parts gives the GEM(1) sequence $V_j = U_j \prod_{i<j}(1-U_i)$ with independent uniform U_i; see [12, Chapter 3]. If $s_j = x_1 + \cdots + x_j$ and $s_0 = 0$, the joint density of its first k terms is $\prod_{j=1}^{k}(1-s_{j-1})^{-1}$ on $s_k < 1$. Conditional on a mass partition, an ordered

tuple of distinct parts $(x_1, \dots, x_k)$ is selected with probability $\prod_{j=1}^{k} x_j/(1-s_{j-1})$. Dividing the former density by this sampling weight gives (A.2). In particular, $\mathbb{E}N_\varepsilon = \log(1/\varepsilon)$ and there is almost surely no part at a fixed positive threshold. □

Proof of Theorem A.1. Put $t = \log(1/\varepsilon)$ and write

$$\Phi(t,w) = \mathbb{E}(1+w)^{N_{e^{-t}}}, \qquad \Psi(t,w) = e^{-wt}\Phi(t,w).$$

For each finite t, $\Phi(t,\cdot)$ is a polynomial, since the threshold count is bounded. By Lemma A.2,

$$\Phi(t,w) = \sum_{k\ge 0} \frac{w^k}{k!} \int_{\substack{x_i > e^{-t} \\ x_1+\cdots+x_k<1}} \prod_{i=1}^{k} \frac{\mathrm{d}x_i}{x_i}. \tag{A.3}$$

The k-fold integral is at most t^k, so $|\Phi(t,w)| \le e^{|w|t}$.

Differentiate the moving lower boundaries in (A.3). Each boundary puts one variable equal to e^{-t}; rescale the remaining variables by $1-e^{-t}$. For $t \ge \log 2$, this gives

$$\partial_t \Phi(t,w) = w\Phi(\log(e^t-1), w). \tag{A.4}$$

The identity holds almost everywhere, and Φ is locally absolutely continuous in t, which suffices below. For $0 \le t < \log 2$, at most one part exceeds e^{-t}, whence $\Phi(t,w) = 1+wt$. Thus for all $t \ge 0$,

$$|\partial_t \Phi(t,w)| \le |w| e^{|w|t} \tag{A.5}$$

almost everywhere.

For $t \ge \log 2$, set $d(t) = -\log(1-e^{-t}) \le 2e^{-t}$. Then $\log(e^t-1) = t - d(t)$. By (A.4)–(A.5),

$$\begin{aligned} \partial_t \Psi(t,w) &= we^{-wt}\{\Phi(t-d(t),w) - \Phi(t,w)\}, \\ |\partial_t \Psi(t,w)| &\le |w|^2 d(t) e^{(|w|-\Re w)t} \le 2\varrho^2 e^{-(1-2\varrho)t}, \qquad |w| \le \varrho < 1/2. \end{aligned} \tag{A.6}$$

This integrable bound proves locally uniform convergence of $\Psi(t,\cdot)$ on $|w| < 1/2$, with error $O_\varrho(e^{-(1-2\varrho)t})$. The limiting function is analytic.

It remains to identify that function. For real $w > 0$, let $\mathcal{N}_{\varepsilon,w}$ be a Poisson process on $[\varepsilon, 1]$ with intensity $w\,\mathrm{d}x/x$, and let $S_{\varepsilon,w}$ be the sum of its points. The Poisson expansion in (A.3) gives

$$\varepsilon^w \mathbb{E}(1+w)^{N_\varepsilon} = \mathbb{P}(S_{\varepsilon,w} < 1). \tag{A.7}$$

As $\varepsilon \downarrow 0$, the sums increase to the time-w Dickman subordinator, with Lévy measure $\mathbf{1}_{(0,1)}(x)\,\mathrm{d}x/x$. Its density on $(0,1]$ is

$$\frac{e^{-\gamma w}}{\Gamma(w)} x^{w-1};$$

see [3, Theorem 1.1]. There is no atom at one, and integration gives $e^{-\gamma w}/\Gamma(1+w)$ for the limit of (A.7). The identity theorem identifies the analytic limit on $|w| < 1/2$. Integrating (A.6) proves the stated rate. □

Proposition A.3 (Small-threshold PD cumulants)**.** *For every fixed $q \ge 2$,*

$$\kappa_q^{\mathrm{F}}(N_\varepsilon) \longrightarrow (-1)^{q+1}(q-1)!\zeta(q). \tag{A.8}$$

More precisely, for every fixed $0 < \varrho < 1/2$ the error is $O_{q,\varrho}(\varepsilon^{1-2\varrho})$. For $q = 2$,

$$\kappa_2^{\mathrm{F}}(N_\varepsilon) = -\int_{\substack{\varepsilon<x,y<1 \\ x+y>1}} \frac{\mathrm{d}x\,\mathrm{d}y}{xy}, \qquad 0 \le \zeta(2) + \kappa_2^{\mathrm{F}}(N_\varepsilon) \le 2\log\frac{1}{1-\varepsilon}. \tag{A.9}$$

Proof. The limiting function in (A.1) is nonzero on $|w| \le \varrho < 1/2$. For sufficiently small ε, take logarithms with value zero at $w = 0$ and use Cauchy's formula. This justifies coefficientwise differentiation, with the asserted error, of

$$\log \mathbb{E}(1+w)^{N_\varepsilon} = w \log(1/\varepsilon) - \gamma w - \log \Gamma(1+w) + O_\varrho(\varepsilon^{1-2\varrho}).$$

Now

$$\log \Gamma(1+w) = -\gamma w + \sum_{q \ge 2} \frac{(-1)^q \zeta(q)}{q} w^q.$$

For $q = 2$, the one- and two-point factorial measures give the integral in (A.9) directly. The integral over the full region $0 < x, y < 1$, $x + y > 1$, equals $\zeta(2)$. The omitted part is at most

$$2 \int_0^\varepsilon \frac{-\log(1-x)}{x} \, \mathrm{d}x \le 2 \int_0^\varepsilon \frac{\mathrm{d}x}{1-x},$$

which proves the final bound. □

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